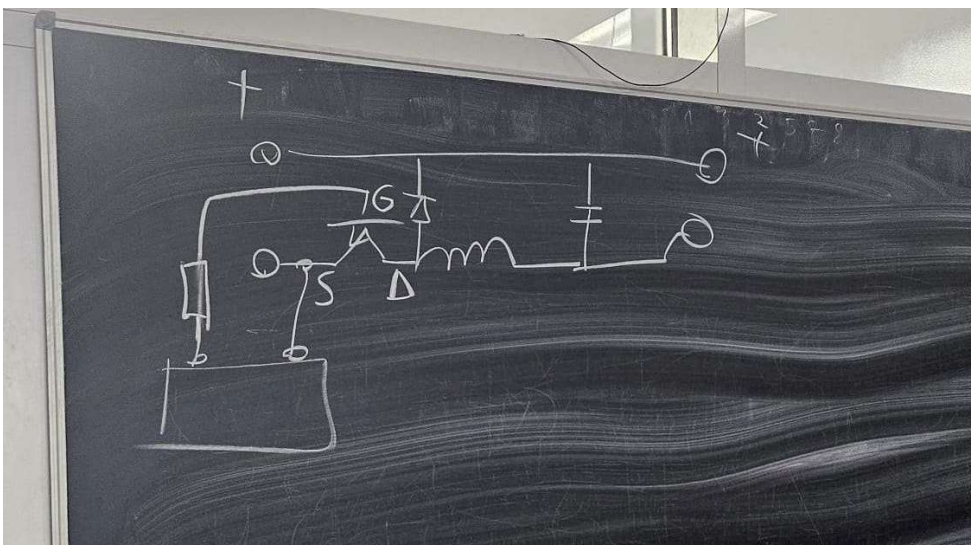
# Buck Convertor

## 1. What is a buck converter and how does it look?

This project covers the steps of making a simple, yet working, buck converter. We first start with what is a buck converter and what are the components that make it:

- It is a type of power electronic circuit that efficiently reduces a higher DC voltage to a lower DC voltage
- To make it we need the next parts:
    - one transistor T
    - one diode D
    - one inductor L
    - one capacitor C




This image shows the schematics of the buck converter and is the main source we use to realize the circuit.

## 2. Calculations

We use formulas from the Project Laboratory to calculate the values of the duty cycle, capacitor, diode and transistor:

$$\mu = \frac{u_{IN}}{u_{OUT}}, \qquad \Delta i_L = \frac{u_{IN} - u_{OUT}}{L} * \frac{\mu}{f}, \qquad \Delta u_C = \frac{\Delta i_L}{8 * f * C},$$

where $\mu$ is the duty cycle and f is the frequency.

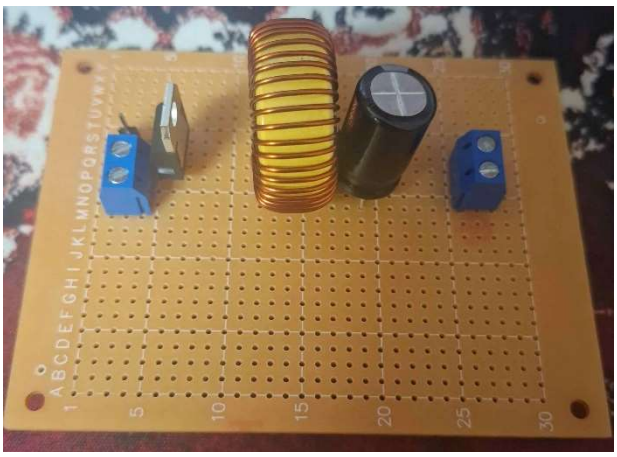
We use the given data: $u_{IN} = 24V, u_{OUT} = 12V, i_{OUT\ MAX} = 3A, \Delta i_L = 40\% * i_{OUT\ MAX} = 1.2A, \Delta u_C = 0.2V$.

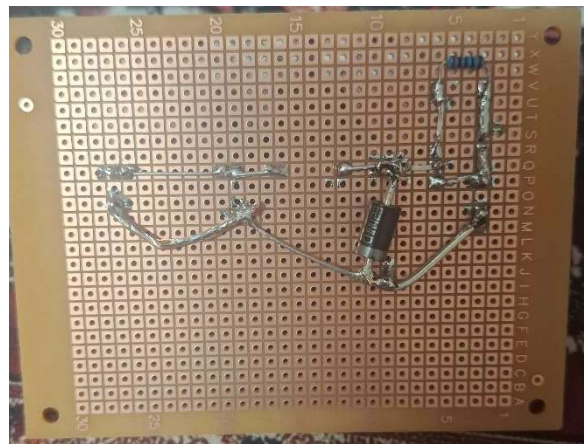
We aim to have: L=0.0001H=100μH, f=50kHz.

After the computations we have: C=0.000015F=15μF, current through T=3.6V and current through D=1.5V.

## 3. The circuit

The second to last part is putting together all the components which we got according to our calculations.

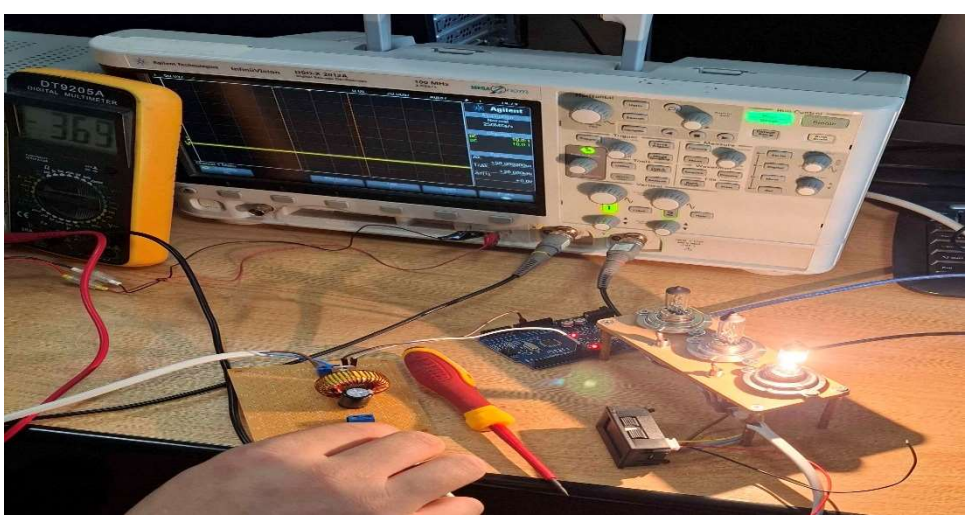




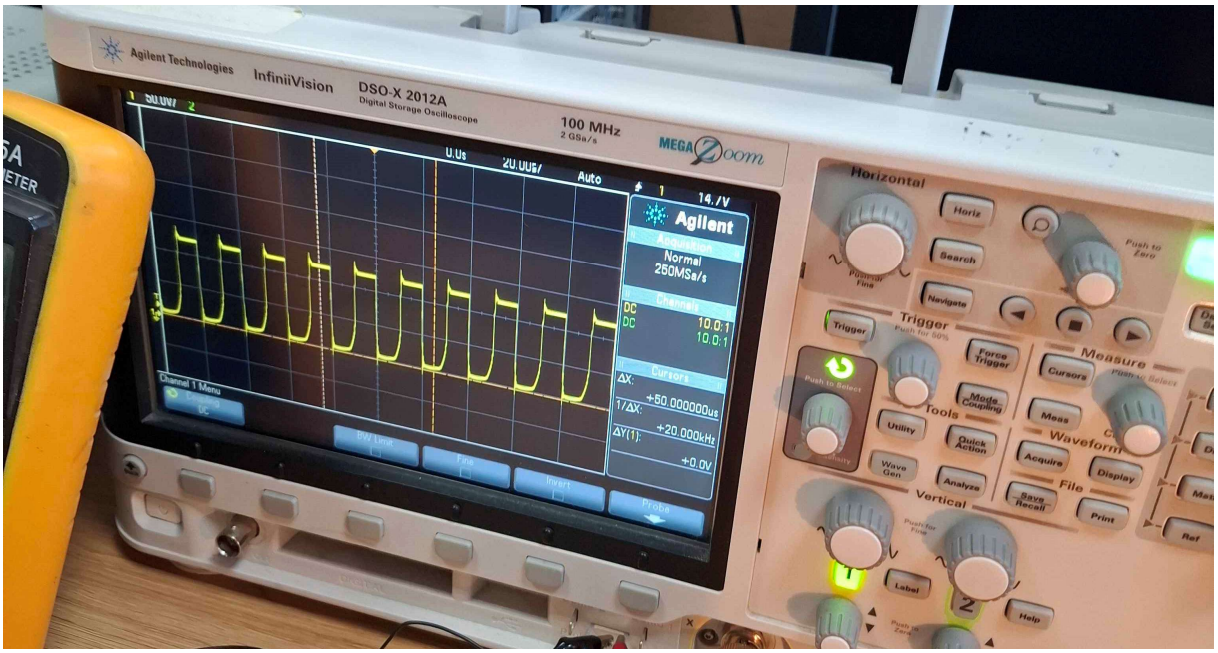
These 2 images show the completed circuit, as well as some additional parts, the resistor so we can connect the converter to an Arduino board without frying it and 2 blue connectors so we can use the converter.

## 4. Testing

The last part of this project is testing the actual circuit and seeing whether it works. We connect the circuit to a DC voltage source, a multimeter and an oscilloscope. We also used an Arduino Uno board to generate the PWM signal.



This first photo shows how to output affects the 3 light bulbs, the power being enough to light only one.




The next photo shows the voltage right after the transistor T.

The last 2 photos show the output voltage, the one on the left being a close up of the one on the right.

## 5. Conclusion

After calculating the values of the components of the buck converter, making and testing the circuit, we can say that the project was a success.